

# Statistical mechanics of the $n$ -queens lattice gas: Monte Carlo simulations and thermodynamic integration

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We investigate the  $n$ -queens problem as a lattice gas—a model in which  $N$  queens are placed on an  $N \times N$  chessboard with pairwise repulsive interactions along shared rows, columns, and diagonals—from the perspective of statistical mechanics. The ground states are exactly the  $Q(N)$  solutions of the classical  $N$ -queens problem, whose entropy per queen  $s_0 \approx \ln N - \gamma$  ( $\gamma \approx 1.944$ ) exhibits a characteristic constraint hierarchy: each successive geometric constraint—columns, then diagonals—reduces the entropy from the free-placement value  $\ln N$  by a definite constant. We derive the exact high-temperature energy  $E/N \rightarrow 5/3$  as  $N \rightarrow \infty$ . Extensive Monte Carlo simulations with  $10^8$  sweeps per temperature point for  $N = 8$ –1024 reveal that the specific heat per queen  $C_v/N$  converges to a universal function of  $T$  as  $N \rightarrow \infty$ , with a non-divergent peak  $C_v^{\max}/N \approx 1.63$  at  $T^* \approx 0.235$  J, establishing the absence of a thermodynamic phase transition. Combined with the trivially exact high-temperature entropy  $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N}$ , the convergence of  $C_v/N$  enables a thermodynamic integration of  $C_v/T$  from  $T = \infty$  to  $T = 0$  that recovers the ground-state entropy—and hence the Simkin constant  $\gamma$ —purely from Monte Carlo data, providing an independent *thermodynamic* route to this fundamental combinatorial constant. Thermodynamic integration yields  $\gamma_{\text{MC}} = 1.946 \pm 0.003$  at  $N = 1024$ , within 0.1% of the precise combinatorial value  $\gamma = 1.944001$ .

## I. INTRODUCTION

The  $N$ -queens problem—placing  $N$  mutually non-attacking queens on an  $N \times N$  chessboard—is among the oldest combinatorial problems in mathematics, dating back to the work of Bezzel in 1848 [1]. Despite its long history, the asymptotic enumeration of solutions was resolved only recently. Simkin [2] proved that the number of solutions  $Q(N)$  satisfies

$$Q(N) \sim \left( \frac{N}{e^\gamma} \right)^N, \quad (1)$$

building on the breakthrough of Bowtell and Keevash [3] and earlier bounds by Luria and Simkin [4]. Simkin’s proof introduces *queenons*—limit objects defined as probability measures on the unit square—and characterizes  $\gamma$  as the solution of a convex optimization problem over this space; the upper bound employs the entropy method while the lower bound is established by a randomized construction algorithm, yielding  $\gamma \in [1.939, 1.945]$ . Nobel, Agrawal, and Boyd [5] subsequently refined these bounds to  $\gamma = 1.944001$  (precision  $\sim 10^{-7}$ ) by solving Simkin’s convex programs with large-scale Newton methods. All of these determinations are purely combinatorial. Equation (1) yields a ground-state entropy per queen  $s_0 = \lim_{N \rightarrow \infty} \ln Q(N)/N = \ln N - \gamma$ .

The form of the ground-state entropy is suggestive. On an  $N \times N$  lattice, a single particle placed freely in one row has  $\ln N$  units of entropy—the maximal combinatorial freedom per particle. The  $N$ -queens entropy  $s_0 = \ln N - \gamma$  can therefore be read as: each queen retains the full lattice entropy  $\ln N$ , reduced by a constant  $\gamma$  that encodes the cumulative cost of geometric constraints. This constant decomposes naturally through

a hierarchy of constraints with increasing stringency [6]:

- *Free placement* (row constraint only):  $\Omega = N^N$ ,  $s = \ln N$ .
- *Permutation* (row + column, also known as the *rook problem* [6]):  $\Omega = N!$ ,  $s \approx \ln N - 1$ .
- *$N$ -queens* (row + column + two diagonals):  $\Omega \approx (N/e^\gamma)^N$ ,  $s \approx \ln N - \gamma$ .

The column constraint costs exactly one unit of entropy (the Stirling correction); the two diagonal constraints together cost  $\gamma - 1 \approx 0.94$  units. That each geometric constraint reduces the entropy by a well-defined constant, independent of  $N$  in the large- $N$  limit, hints at a deeper statistical-mechanical structure underlying the combinatorics.

The additive decomposition of entropy into constraint costs motivates a statistical-mechanical approach. We embed the  $N$ -queens problem in a lattice gas with tunable temperature [7, 8] and perform extensive Monte Carlo simulations. Throughout this paper we set  $k_B = 1$ , so that entropy is dimensionless and temperature has units of energy. A central finding is that the specific heat per queen,  $C_v/N$ , converges to a size-independent curve for sufficiently large  $N$ , exhibiting a Schottky-type peak but no phase transition. Combined with the exactly known high-temperature entropy  $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N} \approx \ln N + 1$ , this convergence enables a thermodynamic integration

$$s_0 = \frac{S(\infty)}{N} - \int_0^\infty \frac{C_v}{NT} dT,$$

which extracts the ground-state entropy—and hence the Simkin constant  $\gamma$ —purely from Monte Carlo simulation data. This provides an independent *thermodynamic*

route to  $\gamma$ : the only combinatorial input is the trivially exact high-temperature entropy  $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N}$ , while the Simkin–Nobel value is used solely for post-hoc comparison. Our thermodynamic extraction yields  $\gamma_{\text{MC}} = 1.946 \pm 0.003$  at  $N = 1024$  (within 0.1% of the precise value [5]), demonstrating that a single Monte Carlo campaign can determine a fundamental combinatorial constant through the laws of thermodynamics.

The paper is organized as follows. Section II defines the model. Section III presents analytical results: the ground-state entropy and constraint hierarchy, and the exact high-temperature limit. Section IV presents Monte Carlo results. Section V carries out the thermodynamic integration. Section VI concludes.

## II. MODEL AND SIMULATION METHODOLOGY

We consider an  $N \times N$  square lattice with open boundary conditions. Each site  $(i, j)$  ( $i, j = 1, \dots, N$ ) carries an occupation variable  $n_{ij} \in \{0, 1\}$ . The Hamiltonian counts mutually attacking queen pairs:

$$H = J \sum_{\langle (i,j), (i',j') \rangle_{\text{attack}}} n_{ij} n_{i'j'}, \quad (2)$$

where the sum runs over all pairs sharing a row ( $i = i'$ ), column ( $j = j'$ ), main diagonal ( $i - j = i' - j'$ ), or anti-diagonal ( $i + j = i' + j'$ ), and  $J > 0$ . We set  $J = 1$  throughout.

The lattice decomposes naturally into *attack lines*:  $N$  rows,  $N$  columns,  $2N - 1$  main diagonals, and  $2N - 1$  anti-diagonals. Writing  $n_\alpha$  for the number of queens on line  $\alpha$ , the Hamiltonian takes the compact form

$$H = \sum_{\alpha \in \mathcal{L}} \binom{n_\alpha}{2}, \quad (3)$$

where  $\mathcal{L}$  denotes the set of all  $6N - 2$  attack lines and  $\binom{n}{2} = n(n-1)/2$  counts the number of attacking pairs on each line. This decomposition into line contributions is the starting point for the high-temperature calculation (Sec. III B).

We work in the canonical ensemble with fixed particle number  $N$  and use Kawasaki dynamics [9] (queen–vacancy exchange) with the Metropolis acceptance criterion [10]. We choose queen–vacancy rather than queen–queen exchanges because the former moves a single queen to any empty site on the board, allowing large spatial displacements in one step, whereas queen–queen swaps merely permute occupied sites and cannot efficiently explore configurations with different spatial structure. One *sweep* consists of  $N$  attempted exchanges. Statistical errors are estimated by jackknife resampling with 200 bins [11]. We simulate eight system sizes  $N = 8, 16, 32, 64, 128, 256, 512, 1024$  on a uniform grid of 280 temperature points spanning  $T = 0.05$  to  $500 J$ , with  $10^8$

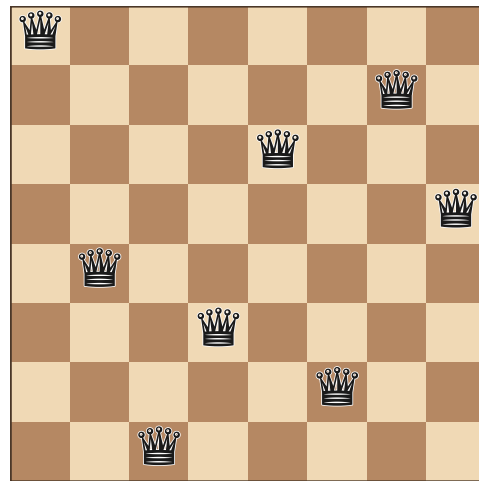


FIG. 1. One of the  $Q(8) = 92$  non-attacking ground-state configurations on an  $8 \times 8$  chessboard. No two queens share a row, column, or diagonal.

measurement sweeps and  $2 \times 10^6$  thermalization sweeps at every temperature point. The grid places 160 points in the specific-heat peak region ( $T = 0.1025$ – $0.500 J$  in steps of  $0.0025 J$ ), 11 points in the low-temperature region ( $T = 0.050$ – $0.100 J$ , step  $0.005 J$ ), and the remaining 109 points at intermediate and high temperatures up to  $T = 500 J$ . All eight sizes share the identical temperature grid and measurement parameters, ensuring a self-consistent data set. The simulations were carried out using 280 CPU cores in parallel, with each core handling all eight system sizes at a single temperature point in serial; the total wall-clock time was approximately nine hours. We verified ergodicity by running independent simulations from multiple random initial configurations (seeds); all runs yielded statistically consistent results for  $E/N$  and  $C_v/N$  at every temperature, confirming that the Kawasaki dynamics explores the configuration space adequately within our thermalization and measurement windows. A representative ground-state configuration is shown in Fig. 1.

## III. ANALYTICAL RESULTS

### A. Ground-state entropy and constraint hierarchy

For  $N$  queens, the ground states ( $E = 0$ ) correspond to the  $Q(N)$  non-attacking configurations. Using Eq. (1), the entropy per queen is

$$s_0 = \lim_{N \rightarrow \infty} \frac{\ln Q(N)}{N} = \ln N - \gamma, \quad \gamma \approx 1.94400. \quad (4)$$

The three-level constraint hierarchy (Table I), implicit in the Stirling approximation and Simkin’s result [2, 6], provides a useful physical interpretation. Starting from

TABLE I. Entropy per queen under successive constraints for the  $N$ -queens problem. The column constraint costs exactly 1 unit; the two diagonal constraints cost  $\gamma - 1 \approx 0.94$  units combined.

| Model       | Constraints        | $\Omega$                 | $s = \frac{1}{N} \ln \Omega$ |
|-------------|--------------------|--------------------------|------------------------------|
| Free        | Row only           | $N^N$                    | $\ln N$                      |
| Permutation | Row + column       | $N!$                     | $\ln N - 1$                  |
| $N$ -queens | Row + col + 2 diag | $\approx (N/e^\gamma)^N$ | $\ln N - \gamma$             |

a fully unconstrained model where each queen independently chooses one of  $N$  columns ( $\Omega_{\text{free}} = N^N$ ,  $s_{\text{free}} = \ln N$ ), the column non-repetition constraint reduces the count to  $N!$  permutations. By Stirling's approximation,  $\ln N! = N \ln N - N + O(\ln N)$ , so the entropy per queen is

$$s_{\text{perm}} = \frac{\ln N!}{N} = \ln N - 1 + O\left(\frac{\ln N}{N}\right). \quad (5)$$

Thus the column constraint costs exactly one unit of entropy—the Stirling correction arising from the indistinguishability of column assignments.

The two diagonal constraints further reduce the entropy to  $s_0 = \ln N - \gamma$ , costing an additional  $\gamma - 1 \approx 0.94$  units, or approximately  $(\gamma - 1)/2 \approx 0.47$  units per diagonal family. The hierarchy of costs is physically natural: the column constraint, being a global permutation on  $N$  equivalent lines, removes exactly one unit of freedom per queen, while each diagonal family—whose lines have unequal lengths ranging from 1 to  $N$ —removes roughly half a unit.

That each geometric constraint reduces the per-queen entropy by a constant independent of  $N$  suggests that the finite-temperature thermodynamics may also admit a simple analytical description. We turn to this question next, beginning with the high-temperature limit.

### B. High-temperature limit

At  $T \rightarrow \infty$ , all  $\binom{N^2}{N}$  configurations are equally probable. Following Polson and Sokolov [8], the exact per-queen energy can be obtained by linearity of expectation over the attack-line decomposition [Eq. (3)]. The probability that any two given sites are simultaneously occupied is  $P = N(N-1)/[N^2(N^2-1)]$ , and the total number of attacking site pairs summed over all  $6N-2$  lines is  $S_{\text{tot}} = N(N-1)(5N-1)/3$ . Combining these gives the exact result

$$\left. \frac{\langle E \rangle}{N} \right|_{T \rightarrow \infty} = \frac{(5N-1)(N-1)}{3N(N+1)} \xrightarrow{N \rightarrow \infty} \frac{5}{3}, \quad (6)$$

valid for all finite  $N$ . Our Monte Carlo simulations (Fig. 3) confirm this: at high  $T$ ,  $E/N$  converges to Eq. (6) with corrections of order  $O(1/N)$ .

## IV. FINITE-TEMPERATURE MONTE CARLO RESULTS

### A. Convergence diagnostics

Before presenting thermodynamic observables, we verify that the Monte Carlo simulations have reached equilibrium and that the measurements are statistically independent. Figure 2 shows the acceptance rate and the integrated autocorrelation time  $\tau_{\text{int}}$  of the energy as functions of temperature for several system sizes.

The acceptance rate [panel (a)] increases monotonically with  $T$ , reaching  $\sim 0.38$ – $0.47$  at  $T = 1 J$ ; it drops below  $10^{-5}$  for  $T \lesssim 0.1 J$ , reflecting the near-perfect freezing into non-attacking ground states. For  $N \geq 32$  the curves collapse, confirming convergence to the thermodynamic limit.

The autocorrelation time [panel (b)] exhibits a sharp peak at  $T \approx 0.055$ – $0.10 J$ , well below the specific heat maximum  $T^* \approx 0.235 J$ , reaching  $\tau_{\text{int}} \approx 5.0 \times 10^4$  sweeps for  $N = 1024$ . Even at this worst case,  $10^8$  measurement sweeps yield  $\sim 2000$  independent samples—adequate for the essentially featureless low-temperature regime where both  $E/N$  and  $C_v/N$  approach zero exponentially. Crucially, at the physically important specific heat peak ( $T \approx 0.235 J$ ),  $\tau_{\text{int}} \approx 85$ – $200$  sweeps for all  $N$ , providing more than  $5 \times 10^5$  independent samples per temperature point and ensuring high statistical precision in the region that matters most. At  $T/J > 0.4$ ,  $\tau_{\text{int}}$  falls to  $\sim 5$ – $10$  sweeps for all system sizes. We additionally verified ergodicity by comparing independent runs from different random initial configurations, all yielding statistically consistent results.

### B. Energy

Figure 3 shows  $E/N$  vs.  $T$  for  $N = 8$ – $1024$ . The curves for  $N \geq 32$  are nearly indistinguishable, confirming rapid convergence to the thermodynamic limit. At high temperatures,  $E/N$  approaches the exact finite- $N$  value [Eq. (6)], which itself converges to  $5/3$  as  $N \rightarrow \infty$ .

### C. Specific heat convergence

The specific heat per queen  $C_v/N = (\langle E^2 \rangle - \langle E \rangle^2)/(T^2 N)$  is shown in Fig. 4. For  $N \geq 32$ , the  $C_v/N$  curves collapse onto a single function of  $T$ , indicating that the thermodynamic limit has been reached. Table II lists the peak values.

The key observation is that  $C_v^{\text{max}}/N$  saturates near 1.63 for  $N \geq 32$  and shows no systematic growth with system size (Table II). The variation across  $N = 32$ – $1024$  is  $\Delta C_v^{\text{max}}/N \lesssim 0.007$ , far smaller than the  $\sim 0.06$  per doubling expected from a logarithmic divergence  $\propto \ln N$ . This behavior is consistent with the absence of a thermodynamic phase transition and identifies the crossover as

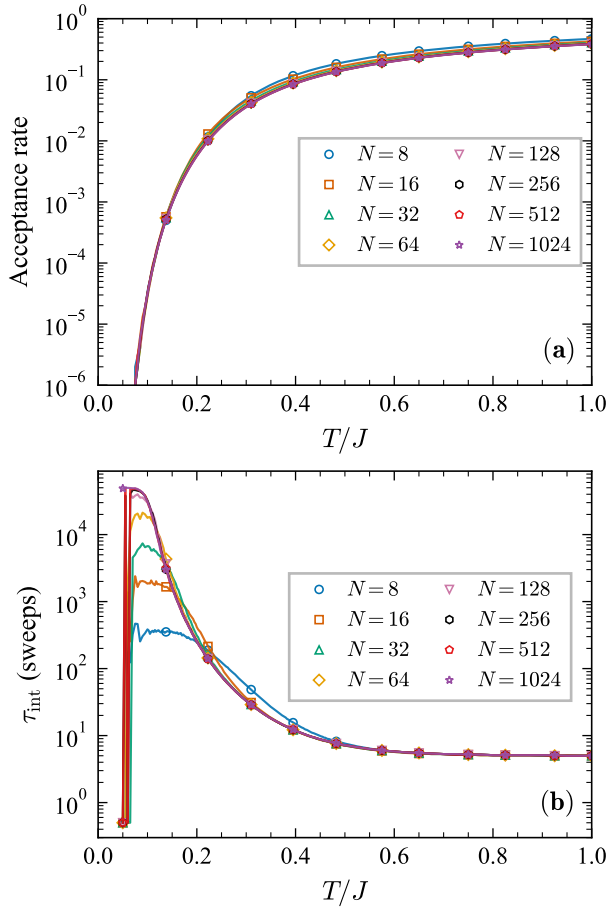


FIG. 2. Convergence diagnostics for the Monte Carlo simulations. (a) Acceptance rate of the Kawasaki (queen–vacancy exchange) moves as a function of temperature for  $N = 8$ –1024 (log scale). The rate drops below  $10^{-5}$  for  $T \lesssim 0.1 J$  and rises monotonically, reaching 0.38–0.47 at  $T = 1 J$ . (b) Integrated autocorrelation time  $\tau_{\text{int}}$  of the energy in units of sweeps (log scale). The sharp peak at  $T \approx 0.055$ – $0.10 J$  reflects low-temperature freezing and lies well below the specific heat maximum  $T^* \approx 0.235 J$ ; at  $T^*$  itself,  $\tau_{\text{int}} \approx 85$ –200 sweeps, yielding more than  $5 \times 10^5$  independent samples from  $10^8$  sweeps.

a Schottky-type anomaly [12]—a broad peak in  $C_v$  arising from thermal excitation across a finite energy gap, without any symmetry breaking. The peak temperature  $T^* \approx 0.235 J$  stabilizes for  $N \geq 64$ .

The convergence of  $C_v/N$  to a definite limiting function has a powerful consequence: since both the high-temperature entropy  $S(\infty)/N$  and the ground-state entropy  $s_0$  are independently known, the integral  $\int_0^\infty C_v/(NT) dT$  must equal their difference. We exploit this in the next section.

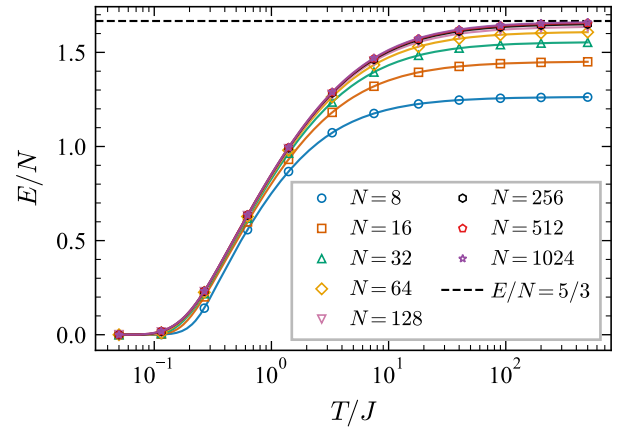


FIG. 3. Energy per queen  $E/N$  vs. temperature (logarithmic scale) for  $N = 8$ –1024 over the full range  $T = 0.05$ – $400 J$ . The dashed line marks the analytical  $N \rightarrow \infty$  high- $T$  limit  $E/N = 5/3$ .

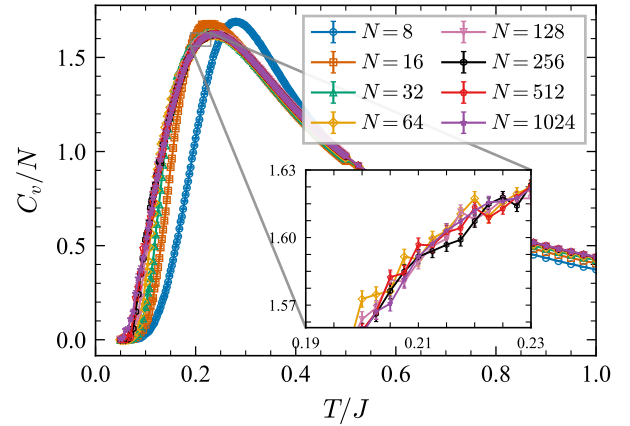


FIG. 4. Specific heat per queen  $C_v/N$  vs.  $T$  for  $N = 8$ –1024 in the peak region ( $T \leq 1 J$ , linear scale) with error bars. Inset: zoom into the peak for  $N \geq 64$ . The curves for  $N \geq 32$  collapse onto a single function, confirming convergence to the thermodynamic limit.

## V. THERMODYNAMIC INTEGRATION AND ENTROPY SELF-CONSISTENCY

The convergence of  $C_v/N$  to a universal function enables a powerful thermodynamic self-consistency check. We compute the entropy difference between  $T = \infty$  and  $T = 0$  in two independent ways and verify that they agree.

*Combinatorial route.*—At  $T \rightarrow \infty$ , all configurations are equally probable, so the entropy per queen is

$$\frac{S(\infty)}{N} = \frac{1}{N} \ln \binom{N^2}{N}. \quad (7)$$

TABLE II.  $C_v/N$  peak scaling. For  $N = 8$ –128, the peak values are from dedicated simulations with dense temperature sampling in  $T \in (0.20, 0.30) J$ ; for  $N = 256$ –1024, values are from the standard 280-point grid. The peak height saturates near 1.63 for  $N \geq 32$ , consistent with the absence of a thermodynamic phase transition.

| $N$  | $T_{\text{peak}}$ | $C_v^{\text{max}}/N$ | error  |
|------|-------------------|----------------------|--------|
| 8    | 0.281             | 1.6915               | 0.0002 |
| 16   | 0.227             | 1.6783               | 0.0006 |
| 32   | 0.231             | 1.6288               | 0.0005 |
| 64   | 0.237             | 1.6216               | 0.0004 |
| 100  | 0.237             | 1.6229               | 0.0004 |
| 128  | 0.237             | 1.6225               | 0.0004 |
| 256  | 0.2450            | 1.6238               | 0.0020 |
| 512  | 0.2400            | 1.6287               | 0.0023 |
| 1024 | 0.2325            | 1.6251               | 0.0024 |

For  $N \ll N^2$ , applying the Stirling approximation  $\ln n! = n \ln n - n + \frac{1}{2} \ln(2\pi n) + O(n^{-1})$ :

$$\begin{aligned} \ln \binom{N^2}{N} &= \ln(N^2!) - \ln N! - \ln(N^2 - N)! \\ &= N \ln N - (N^2 - N) \ln(1 - 1/N) + \Delta_S, \end{aligned} \quad (8)$$

where the first two terms come from the  $n \ln n - n$  part of Stirling's formula and the correction  $\Delta_S = \frac{1}{2} \ln[N^2/(2\pi N(N^2 - N))] = -\frac{1}{2} \ln(2\pi(N - 1))$  collects the subleading logarithms. Expanding  $\ln(1 - 1/N) = -1/N - 1/(2N^2) - \dots$ , the leading terms give  $-(N^2 - N) \ln(1 - 1/N) = N - \frac{1}{2} + O(N^{-1})$ , so

$$\frac{S(\infty)}{N} = \ln N + 1 + O\left(\frac{\ln N}{N}\right). \quad (9)$$

At  $T = 0$ , only ground states contribute:  $S(0)/N = \ln Q(N)/N \approx \ln N - \gamma$ .

*Thermodynamic route.*—The fundamental relation

$$S(\infty) - S(0) = \int_0^\infty \frac{C_v}{T} dT \quad (10)$$

relates the entropy difference to the specific heat. Dividing by  $N$  and substituting the results above:

$$\int_0^\infty \frac{C_v}{NT} dT = \frac{S(\infty) - S(0)}{N} = 1 + \gamma + O\left(\frac{\ln N}{N}\right). \quad (11)$$

This is a remarkable relation: the left side involves  $C_v(T)$  measured by simulation over the entire temperature range, while the right side involves only the *combinatorial* constants 1 (from the Stirling correction) and  $\gamma$  (from the Simkin asymptotic formula).

Rearranging Eq. (11), we can extract the ground-state entropy per queen directly from the Monte Carlo data:

$$s_0^{\text{MC}} = \frac{S(\infty)}{N} - \int_0^\infty \frac{C_v}{NT} dT, \quad (12)$$

where  $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N}$  is the trivially exact entropy of the uniform distribution over all  $\binom{N^2}{N}$  configurations—requiring no knowledge of the ground-state count  $Q(N)$  or the Simkin constant. This provides a purely thermodynamic determination of the ground-state degeneracy without direct enumeration; the Simkin–Nobel value  $\gamma = 1.944001$  enters only as a benchmark for comparison.

The numerical integration uses 280 temperature grid points spanning from  $T = 0.05 J$  to  $400 J$ ; the low- $T$  cutoff contributes negligibly since  $C_v/T$  is exponentially suppressed below  $T \approx 0.1 J$ , and the high- $T$  cutoff at  $400 J$  is sufficient because  $C_v/T \propto T^{-3}$  decays rapidly, contributing less than 0.01% of the integral beyond this point. The trapezoidal rule on the logarithmic grid introduces discretization errors well below the statistical uncertainty of the Monte Carlo data.

Table III summarizes the results. For  $N = 8$  and  $N = 16$ , where exact values of  $Q(N)$  are known ( $Q(8) = 92$ ,  $Q(16) = 14\,772\,512$ ),  $s_0^{\text{MC}}$  agrees with the exact  $s_0^{\text{exact}} = \ln Q(N)/N$  to within 0.1%, validating the simulation and integration procedure.

For  $N \geq 32$ , where exact  $Q(N)$  is unknown, we use  $s_0^{\text{MC}}$  to extract the Simkin constant via  $\gamma_{\text{MC}} = \ln N - s_0^{\text{MC}}$ . The extracted values converge toward the known asymptotic value  $\gamma = 1.944001$  [5]:  $\gamma_{\text{MC}} = 1.833 \pm 0.001$  ( $N = 32$ ),  $1.885 \pm 0.002$  ( $N = 64$ ),  $1.909 \pm 0.002$  ( $N = 128$ ),  $1.925 \pm 0.002$  ( $N = 256$ ),  $1.931 \pm 0.003$  ( $N = 512$ ), and  $1.946 \pm 0.003$  ( $N = 1024$ ), with the deviation decreasing from 5.7% to 0.11% at  $N = 1024$ . The dominant source of error at moderate  $N$  is the finite-size correction to the Simkin formula, which makes  $\gamma_{\text{eff}}(N) = \ln N - s_0(N)$  systematically smaller than the asymptotic  $\gamma$ ; the MC integration itself introduces only sub-percent errors, as confirmed by the  $N = 8$  and  $N = 16$  benchmarks. The statistical uncertainties on  $s_0^{\text{MC}}$  and  $\gamma_{\text{MC}}$  (Table III) are obtained by propagating jackknife errors on the individual  $C_v(T)$  data points through the trapezoidal integration. At  $N = 1024$ , the propagated uncertainty  $\sigma(\gamma_{\text{MC}}) = 0.003$  is small compared to the 0.11% deviation from the asymptotic value, confirming that the dominant source of error is the finite-size correction to the Simkin formula rather than statistical noise.

A finite-size scaling extrapolation (e.g.,  $\gamma_{\text{MC}}(N) = \gamma + a N^{-\alpha}$ ) could in principle improve the infinite- $N$  estimate. However, the functional form of the subleading corrections to  $Q(N)$  is not known analytically, making any assumed fitting form ad hoc. We therefore report the raw  $\gamma_{\text{MC}}(N)$  values without extrapolation and note that extending the simulations to larger  $N$  would further narrow the gap.

The monotonic convergence of  $\gamma_{\text{MC}}$  toward  $\gamma = 1.944001$  with increasing  $N$ —reaching 0.11% accuracy at  $N = 1024$ —confirms the mutual consistency of the Monte Carlo thermodynamics, the Stirling approximation for  $S(\infty)$ , and the Simkin combinatorial formula.

TABLE III. Ground-state entropy from thermodynamic integration.  $s_0^{\text{MC}} = S(\infty)/N - \int_0^\infty (C_v/NT) dT$ , where  $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N}$  is exact. Statistical uncertainties are obtained by propagating jackknife errors on  $C_v(T)$  through the trapezoidal integration. For  $N \leq 16$ ,  $s_0^{\text{MC}}$  is compared with the exact ground-state entropy  $s_0^{\text{exact}} = \ln Q(N)/N$ . For  $N \geq 32$ , we extract  $\gamma_{\text{MC}} = \ln N - s_0^{\text{MC}}$  and compare with the precise value  $\gamma = 1.944001$  [5]. All data use a 280-point temperature grid with  $10^8$  measurement sweeps per point.

| $N$  | $s_0^{\text{MC}}$ | $s_0^{\text{exact}}$ | $\gamma_{\text{MC}}$ | Deviation |
|------|-------------------|----------------------|----------------------|-----------|
| 8    | $0.566 \pm 0.000$ | 0.565                | —                    | 0.1%      |
| 16   | $1.031 \pm 0.001$ | 1.032                | —                    | 0.1%      |
| 32   | $1.633 \pm 0.001$ | —                    | $1.833 \pm 0.001$    | 5.7%      |
| 64   | $2.274 \pm 0.002$ | —                    | $1.885 \pm 0.002$    | 3.0%      |
| 128  | $2.943 \pm 0.002$ | —                    | $1.909 \pm 0.002$    | 1.8%      |
| 256  | $3.620 \pm 0.002$ | —                    | $1.925 \pm 0.002$    | 1.0%      |
| 512  | $4.308 \pm 0.003$ | —                    | $1.931 \pm 0.003$    | 0.69%     |
| 1024 | $4.985 \pm 0.003$ | —                    | $1.946 \pm 0.003$    | 0.11%     |

## VI. CONCLUSIONS

We have demonstrated that the Simkin constant  $\gamma$ —a fundamental quantity in combinatorics—can be determined through an independent thermodynamic route. By formulating the  $n$ -queens problem as a lattice gas and performing extensive Monte Carlo simulations ( $N = 8$ – $1024$ ,  $10^8$  sweeps per temperature point), we map out the specific heat  $C_v/N$  over the entire temperature range. A key prerequisite is that  $C_v/N$  converges to a size-independent function for  $N \geq 32$ , with a non-divergent Schottky-type peak at  $T^* \approx 0.235 J$  consistent with the absence of a thermodynamic phase transition.

This convergence, combined with the trivially exact high-temperature entropy  $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N}$ , enables a thermodynamic integration of  $C_v/T$  that yields the Simkin constant as  $\gamma_{\text{MC}} = \ln N - S(\infty)/N + \int_0^\infty (C_v/NT) dT$ . The method is validated at  $N = 8$  and  $16$  where exact solution counts are known (agreement within 0.1%), and for larger  $N$  produces  $\gamma_{\text{MC}} = 1.946 \pm 0.003$  at  $N = 1024$ , within 0.11% of the precise combinatorial value  $\gamma = 1.944001$  [5]. The remaining deviation is dominated by finite-size corrections to the Simkin formula rather than statistical noise, and extending to still larger system sizes would further sharpen the estimate.

These results establish the  $n$ -queens lattice gas as a setting where combinatorial and thermodynamic information are connected quantitatively: a single Monte Carlo campaign suffices to determine a fundamental combinatorial constant through the laws of thermodynamics.

## VII. DATA AVAILABILITY

The simulation data and analysis code are publicly available on GitHub [13].

## VIII. ACKNOWLEDGMENTS

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- [1] M. Bezzel, Schachfreund, Berliner Schachzeitung **3**, 636 (1848).
  - [2] M. Simkin, The number of  $n$ -queens configurations, Adv. Math. **427**, 109127 (2023).
  - [3] C. Bowtell and P. Keevash, The  $n$ -queens problem, arXiv:2109.08083 (2021).
  - [4] Z. Luria and M. Simkin, A lower bound for the  $n$ -queens problem, arXiv:2105.11431 (2021).
  - [5] P. Nobel, A. Agrawal, and S. Boyd, Computing tighter bounds on the  $n$ -queens constant via Newton’s method, Optim. Lett. **17**, 1229–1240 (2023).
  - [6] D. E. Knuth, *The Art of Computer Programming*, Vol. 4B (Addison-Wesley, 2022), Sec. 7.2.2.
  - [7] C. Zhang and J. Ma, Counting solutions for the  $N$ -queens and Latin-square problems by efficient Monte Carlo simulations, Phys. Rev. E **79**, 016703 (2009).
  - [8] N. Polson and V. Sokolov, Counting  $N$  queens, arXiv:2407.08830 (2024).
  - [9] K. Kawasaki, Diffusion constants near the critical point for time-dependent Ising models. I, Phys. Rev. **145**, 224 (1966).
  - [10] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, Equation of state calculations by fast computing machines, J. Chem. Phys. **21**, 1087 (1953).
  - [11] B. Efron, *The Jackknife, the Bootstrap, and Other Resampling Plans* (SIAM, Philadelphia, 1982).
  - [12] K. Binder and D. W. Heermann, *Monte Carlo Simulation in Statistical Physics* (Springer, Berlin, 5th ed., 2010).
  - [13] Z.-Y. Liu and L. Wang, Simulation code and data for the  $n$ -queens lattice gas, <https://github.com/LiuZY613/nqueen-lattice-gas>.